

OPEN-HIRE – A Freelancing Website Server that provides projects & GET Hired

A DESIGN THINKING PROJECT REPORT

Submitted by

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AICTE Approved; UGC Autonomous; JNTUH Affiliated; UGC “College with Potential for Excellence”; NAAC A++” Grade
ISO 9001:2015 Certified; QS I.GAUGE “Diamond” Rated; NIRF 2020: 127th Rank Engineering (151-200 Band Overall)

NBA Accredited: CE, CSE, ECE, EEE, EIE, IT, ME; JNTUH -Recognised Research Centres: CE, CSE, ECE, EEE, ME
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2024-2025

**VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI
INSTITUTE OF ENGINEERING AND TECHNOLOGY
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CERTIFICATE

This is to certify that the project titled **"OPEN-HIRE"** is being submitted, by **Aishwarya Nellutla (23071A6901, CSE-IOT), Abhideep Bonigala (23071A66D2, CSE-AIML and Akshitha J (24075A1009, EIE), Bollavarapu Aashika (22071A6771, CSE-DS)** in partial fulfilment of the requirement for the award of degree of **Bachelor of Technology** to the Centre for Presence and Design Thinking at the **Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology** is a record of bonafide work carried out by them under our pedagogy. The results embodied in this Project have not been submitted to any other University or Institute for the award of any degree.

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External Examiners

ABSTRACT

Open Hire is an innovative digital recruitment platform designed to bridge the gap between job seekers and employers through an AI-powered, transparent, and skill-based hiring system. This project presents a structured approach to modern hiring, addressing inefficiencies in traditional recruitment processes and offering a seamless experience for both candidates and recruiters.

The initial chapters focus on identifying the challenges in current hiring practices, including bias, lack of accessibility, and inefficiency. Subsequent chapters introduce the system design and architecture of Open Hire, detailing the user interface, backend integration, and database management. The platform uses machine learning algorithms to match candidates with job profiles based on skills, experience, and preferences, while also offering automated screening tools such as resume parsing, ranking, and real-time interview scheduling.

Further chapters discuss implementation strategies, the development lifecycle using agile methodology, and the integration of communication features like chat and notifications. The final chapters evaluate system performance, user feedback, and scalability, highlighting future enhancements such as remote assessment tools and analytics dashboards.

Through this project, Open Hire demonstrates a scalable, user-friendly, and efficient solution to modern hiring challenges, aiming to democratize opportunities and foster merit-based employment across industries

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CHAPTER 1

INTRODUCTION

1.1 Objective

Open Hire is a smart recruitment platform aimed at transforming the traditional hiring process through technology-driven solutions. It connects job seekers and employers on a single platform, ensuring transparent, skill-based, and unbiased hiring. The ultimate goal of Open Hire is to create a fair and efficient hiring ecosystem that enhances accessibility and merit-based employment for all.

1.2 Introduction

The recruitment industry has long faced challenges such as biased hiring, inefficiencies in screening processes, and limited access to opportunities for deserving candidates. Open Hire addresses these issues by providing an AI-powered recruitment platform that automates key hiring tasks such as resume parsing, skill-based matching, and candidate shortlisting. The system is built using modern web technologies and follows an agile development methodology, focusing on usability, scalability, and performance.

1.3 Motivation

The current hiring process is often biased, time-consuming, and inefficient, with over 85% of applications rejected without proper review. *Open Hire* addresses these issues by offering an AI-driven recruitment platform that ensures fair, skill-based, and transparent hiring. It automates resume screening, matches candidates to job roles based on merit, and eliminates manual errors and unconscious bias. This project not only improves organizational efficiency but also supports a more inclusive and diverse workforce, positively impacting society, technology adoption, and the modern employment landscape.

1.4 Scope for the Work

However, the scope of this project does **not** include advanced features such as real-time video interviews, deep behavioural analysis, integration with external job portals, or mobile application development. This work is intended as a prototype-level solution, showcasing the core functionalities of an intelligent hiring platform. Future expansions may consider the inclusion of advanced analytics, interview scheduling tools, and integration with comprehensive HR management systems for a full-cycle recruitment solution.

CHAPTER 2

DISCOVER AND DEFINE

2.1 Empathy Survey

To better understand user behaviour and pain points, we conducted empathy interviews using Google Forms. This tool allowed us to reach a broader audience remotely and collect qualitative insights in a structured manner. The survey was designed to gather user experiences, preferences, challenges, and expectations.

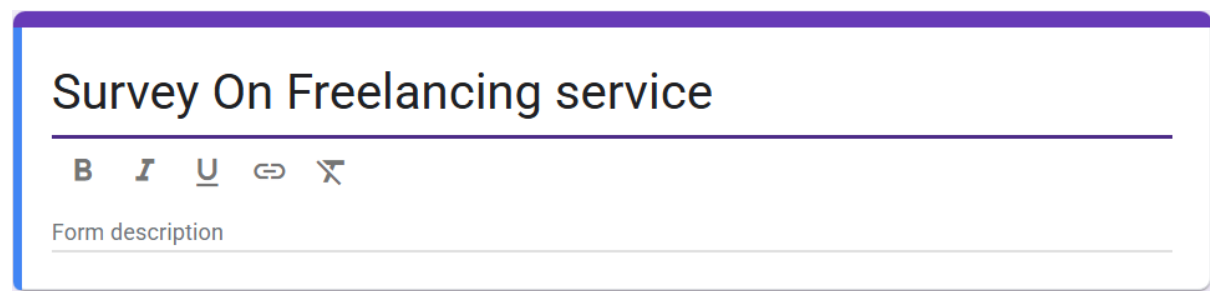
A screenshot of a Google Form titled "Survey On Freelancing service". The form has a purple header bar. Below the title, there is a text area with formatting icons (B, I, U, link, unlink) and a "Form description" label. The form is currently empty.

Image of Empathy Type - Survey

2.1.1 Sample Questions

Based on the survey responses, we created user personas to represent our key audience segments. These personas help us empathize with our users and guide design decisions by highlighting their needs, motivations, and pain points.

- Did you ever face problems in finding a project?
- Did you have experience in contributing to Open - Source Contribution?
- Do you know about these contributions?
- Do you feel these freelancing opportunities are required for the present job market?
- Do you feel freelancing projects are recognised in this present market of high competitions?

2.3 User Needs

Primary Needs:

- Users need a verified platform to avoid scams and fake job posts
- Users want a portfolio section to showcase their work directly
- They expect transparent communication channels with recruiters/employers

Secondary Needs

- There is a strong desire for **project-based filters** tailored to user skills

CHAPTER 3

CUSTOMER SERVICE EXPERIENCE

3.1 Service Experience Cycle

The service experience cycle represents the complete journey a user undergoes while interacting with a platform — from the initial point of contact, through active engagement, to post-service support. Understanding and optimizing each stage of this cycle is crucial for creating a seamless, reliable, and user-centric experience. A well-designed service experience ensures that user expectations are met consistently, enhancing satisfaction, trust, and long-term engagement.

3.2 Pain Point Address

One of the most significant challenges users face is the prevalence of scams and fake job postings. Many users reported that after applying to seemingly legitimate opportunities, they were later asked to pay fees or share sensitive information. This lack of trust stems from the platform's inability to verify recruiters and job postings, which results in users feeling unsafe and hesitant to engage with new opportunities. It undermines the credibility of the platform and creates a major barrier for genuine users seeking reliable work.

Another key issue is the absence of personalized project discovery. Users frequently expressed frustration over having to sift through irrelevant job listings that do not match their skill set or interests. Without project-based filters or intelligent matching systems, the search process becomes time-consuming and inefficient. This reduces user engagement and leads to dissatisfaction, especially among users looking for freelance or skill-specific work.

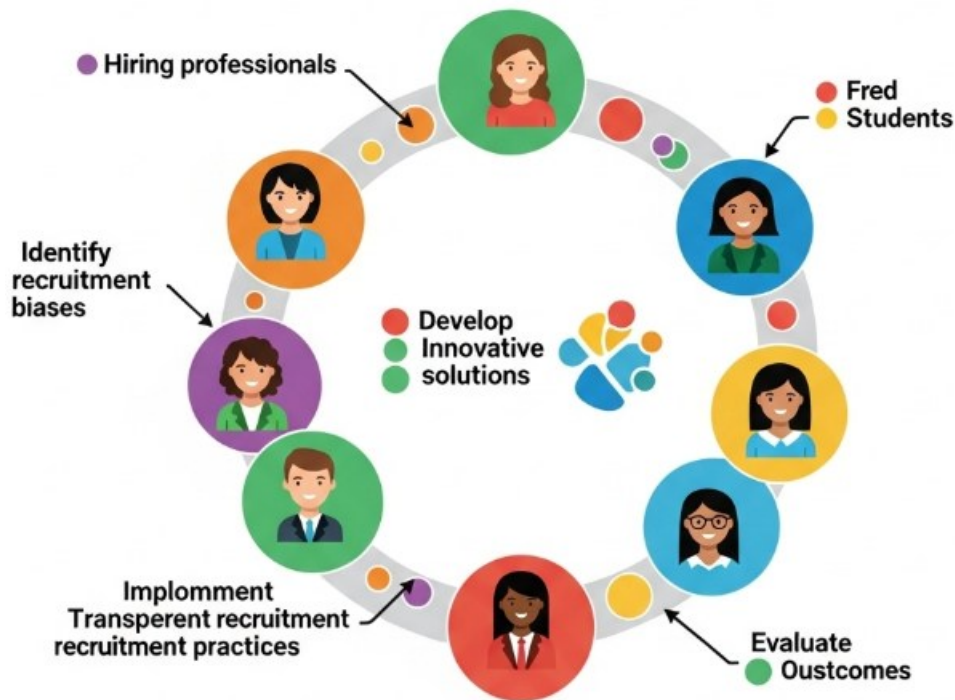
Additionally, users highlighted the lack of a dedicated space to showcase their portfolios. Many felt that without the ability to display their previous work, it became difficult to attract potential employers or stand out among applicants. This limitation affects their professional visibility and reduces their chances of being hired for relevant roles.

CHAPTER 4

IDEATION

4.1 Multiplication-ideation technique :

Open hire unites hiring professionals and students to tackle recruitment challenges through diverse perspectives. This collaboration multiplies insights, drives innovation, and improves hiring practices. By fostering transparency, inclusivity, and shared knowledge, Open hire bridges the gap between job seekers and employers, creating a more efficient and equitable recruitment ecosystem for everyone involved.



Multiplication Process for Open Hire

4.3 Empathy Mapping

SAYS

“I want to contribute to open source, but I don’t know where to start.”

Users express frustration over finding beginner-friendly projects and recognition for their work. They often say GitHub is overwhelming and that recruiters overlook real coding efforts unless showcased through portfolios or direct experience.

THINKS

“I hope my contributions will help me land a job.”

Users think about their career growth and how they can gain real-world experience through open source. They often worry about whether their efforts will be noticed, and if they’re spending time on the right projects to boost their employability.

DOES

Spends hours browsing GitHub, reading documentation, and attempting pull requests.

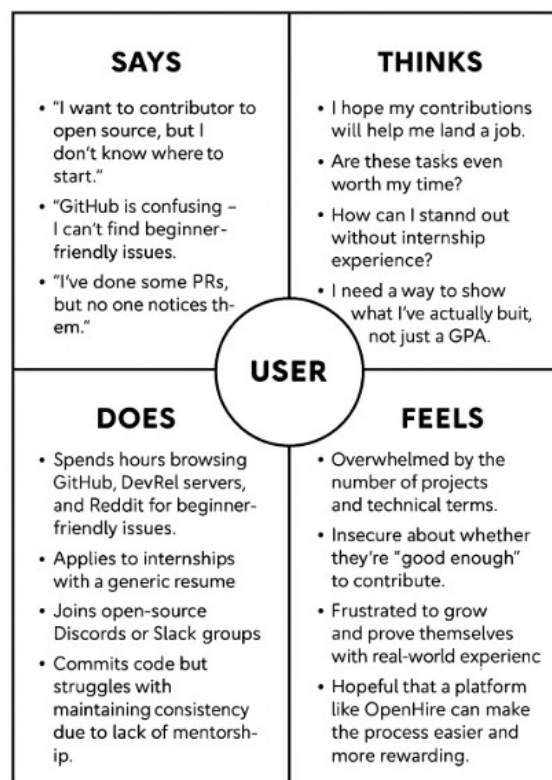
Users frequently switch between projects, apply to internships, and manually update resumes. They join communities for help but often abandon contributions due to unclear onboarding or lack of feedback from maintainers, slowing down their learning and growth journey.

FEELS

Overwhelmed, underconfident, yet motivated.

Users feel excited about contributing to open source but also anxious due to lack of guidance or visible progress. They want to prove themselves but feel lost in the crowd, hoping for a platform that recognizes and supports their journey toward becoming a professional developer.

EMPATHY MAP



Empathy Map

4.2 Outcome of Ideation Phase

The Ideation phase in design thinking is about generating a broad set of creative solutions based on the insights defined in the earlier phases. When you're focused on transforming interview preparation using AI, the outcome of the ideation phase includes a set of innovative, AI-powered concepts or features that can improve how candidates or interviewers prepare for interviews

CHAPTER 5

PROTOTYPE MODEL

5.1 Latent Needs Addressed

1. Progress Transparency & Feedback Loop

- **Need**
Users may not openly ask for detailed updates, but they feel anxious and disconnected when there's no visibility into their application status.
- **Our Solution**
We introduced a **real-time application status tracker** and a **feedback mechanism** that keeps users informed at every stage—applied, under review, selected, or rejected—providing transparency and improving trust in the platform.

2. Personalized Opportunity Matching

- **Need**
Users want job suggestions that match their skills but may not articulate the underlying desire for a personalized experience without manual filtering.
- **Our Solution**
We built an AI-driven skill tagging and job matching system that automatically recommends projects and roles tailored to the user's profile and skill set, streamlining the discovery process.

5.2 Evaluation of prototype Based on Desirability, Feasibility& Viability

Evaluating Open Hire through the Desirability, Feasibility, and Viability (DFV) framework helps assess the platform's potential success in the competitive job-tech and freelancing market. The prototype is designed to support job seekers, freelancers, and verified employers through a secure, intuitive, and skill-aligned experience.

5.2.1 Desirability

This dimension evaluates whether Open Hire effectively fulfills real user needs and creates an engaging, inclusive experience for job seekers and project-based professionals.

5.2.1.1 User Needs

Open Hire addresses core user demands such as authentic job listings, project-based filtering, portfolio showcasing, and clear communication with recruiters. These features support users in building their careers with confidence and clarity, particularly those seeking short-term, skill-specific roles.

5.2.1.2 Customer Experience

The platform delivers a seamless experience through an intuitive dashboard interface, real-time application tracking, personalized job suggestions, and in-app chat support. These reduce the cognitive load and increase the satisfaction of users navigating job opportunities.

5.2.1.3 Inclusivity and Accessibility

Open Hire is inclusive by design—featuring a clean, mobile-first interface, accessible fonts, and easy onboarding for users of various digital literacy levels. It supports underrepresented professionals with equitable access to freelance and employment opportunities.

5.2.1.4 Market Demand

In an era where users seek reliable, skill-oriented job platforms, Open Hire fills a gap by offering verified listings, skill-mapped opportunities, and transparent workflows. Rising demand for micro-jobs and freelance projects positions Open Hire as a timely and relevant solution.

5.2.2 Feasibility

Feasibility analyses whether Open Hire is practical to build, deploy, and scale using current tools, technologies, and infrastructure.

5.2.2.1 Technical Feasibility

The prototype is built on the MERN stack (MongoDB, Express.js, React, Node.js), ensuring scalability and responsiveness. Tools like Firebase (for auth and notifications) and Cloudinary (for file uploads) further enhance its functionality.

5.2.2.2 Integration with Existing Systems

Open Hire is designed to easily integrate with LinkedIn, GitHub, or Google sign-in, allowing streamlined profile creation. Employer systems can use open APIs to connect with applicant tracking systems (ATS) or HR dashboards.

5.2.2.3 Scalability

Its modular architecture and cloud-native infrastructure ensure scalability across regions, institutions, or startups. As demand increases, Open Hire can onboard more recruiters, professionals, and institutions without performance loss.

5.2.2.4 Regulatory Compliance

The platform follows data protection regulations (GDPR, IT Act, etc.) with encrypted user data, consent-based access, and privacy toggles. It ensures user confidence through responsible handling of resumes and portfolios.

5.2.2.5 Cost of Implementation

Development costs are kept low through the use of open-source frameworks and free-tier tools. With minimal DevOps requirements and the possibility of cloud credits or startup funding, Open Hire is both affordable and maintainable.

5.2.3 Viability

Viability examines whether Open Hire is sustainable financially and whether it has room to grow in its domain over time.

5.2.3.1 Economic Viability

Monetization opportunities include **premium employer plans, resume review services, portfolio enhancement tools, and sponsored job listings**. These provide recurring revenue while keeping core features free for job seekers.

5.2.3.2 Market Opportunity

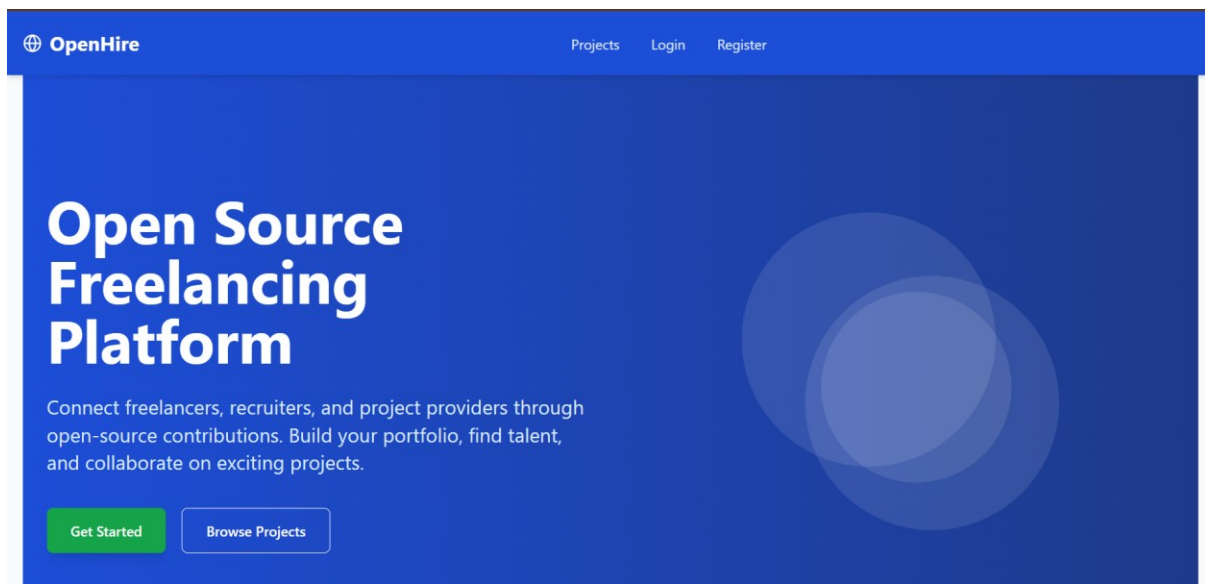
With an increase in **remote work, freelancing, and gig-based employment**, platforms like Open Hire are in high demand. Institutions and startups alike are looking for **curated, skill-matched** talent pools, which Open Hire offers.

5.2.3.3 Business Model

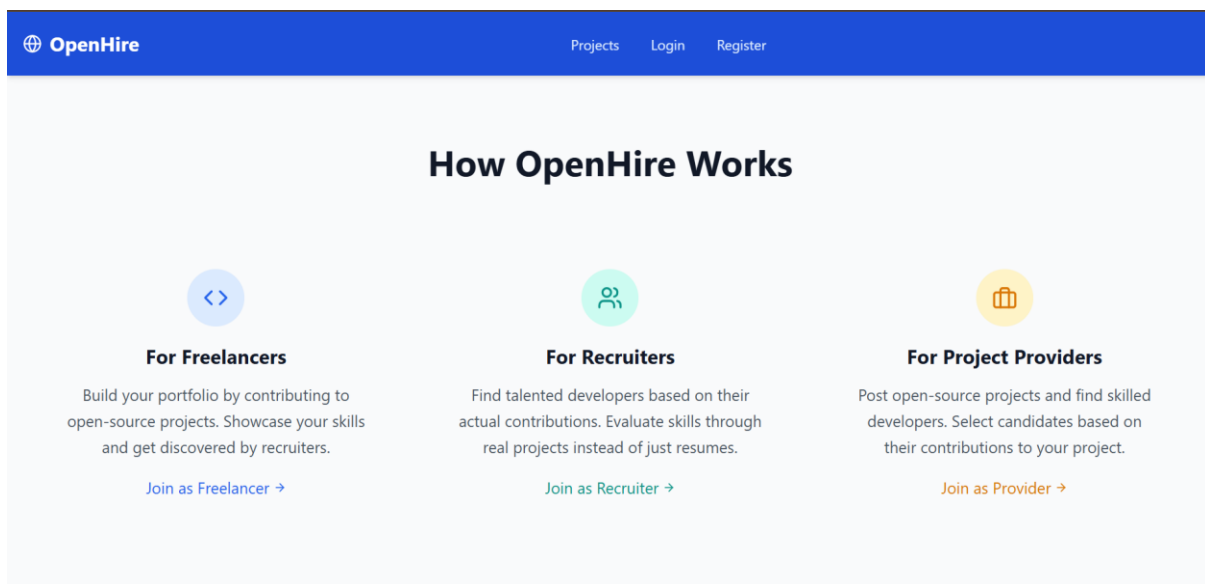
Open Hire can adopt a **freemium SaaS model** for recruiters and institutions, offering analytics dashboards, job performance insights, and skill demand tracking. Students and professionals can opt for premium features like profile audits or faster applications.

5.2.3.4 Competition

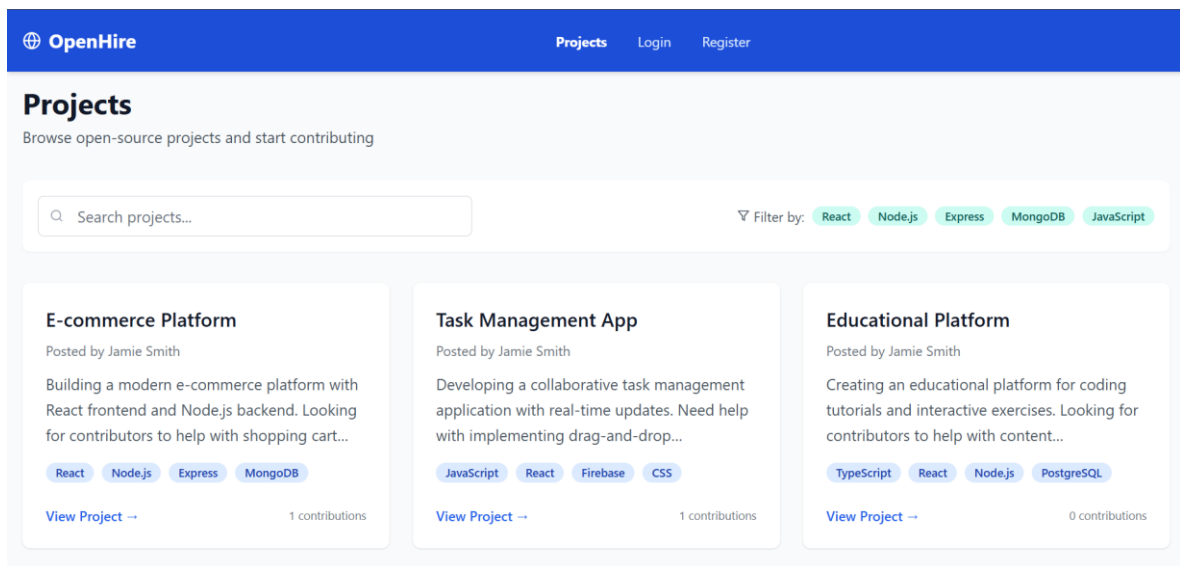
Unlike platforms like **Freelancer** or **Github**, Open Hire uniquely focuses on **verified, project-based, and skill-filtered** jobs. Its strength lies in simplicity, speed, and trust—offering a competitive edge through personalized discovery, verified listings, and a clean user experience.



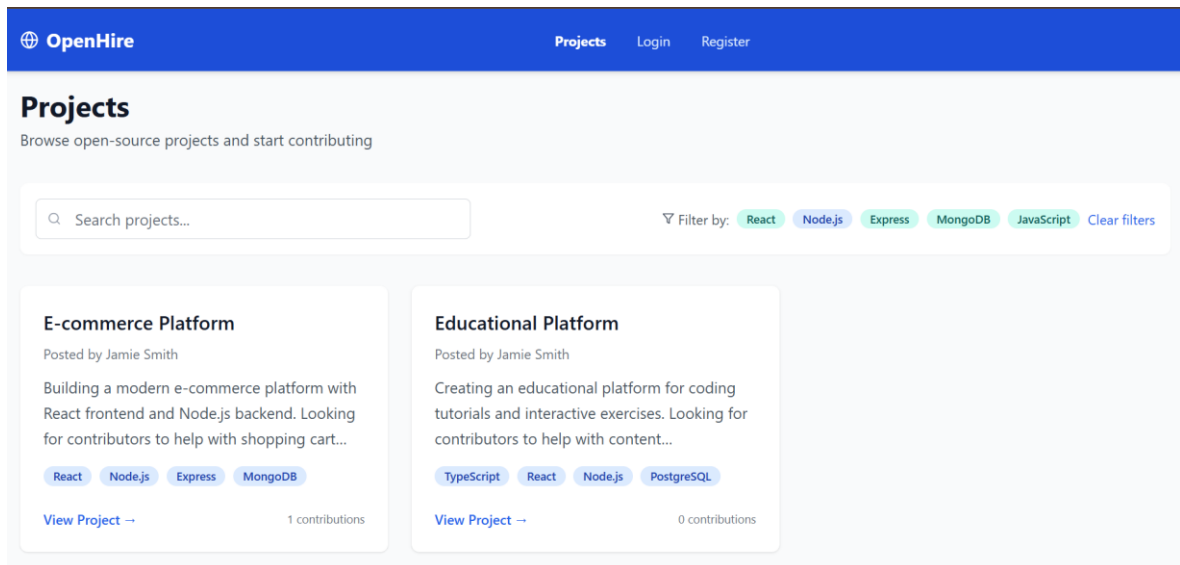
Dashboard for OPEN HIRE Home Page



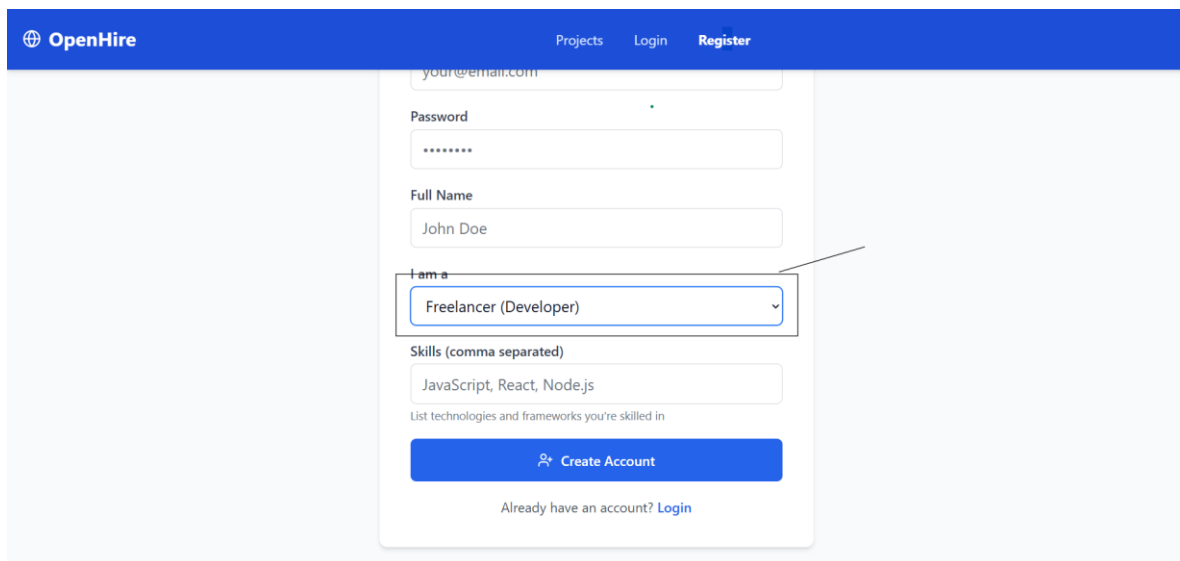
Allows 3 different types of Users



Projects Home Page



Applying Filters and Migrating among projects



The registration form for OpenHire includes a blue header with the logo and navigation links. The form fields are: Email (placeholder: your@email.com), Password (masked with dots), Full Name (John Doe), a dropdown menu for 'I am a' (selected: Freelancer (Developer)), and Skills (comma separated: JavaScript, React, Node.js). A blue 'Create Account' button is at the bottom, with a 'Login' link for existing users.

OpenHire

Projects Login Register

your@email.com

Password

Full Name

John Doe

I am a

Freelancer (Developer)

Skills (comma separated)

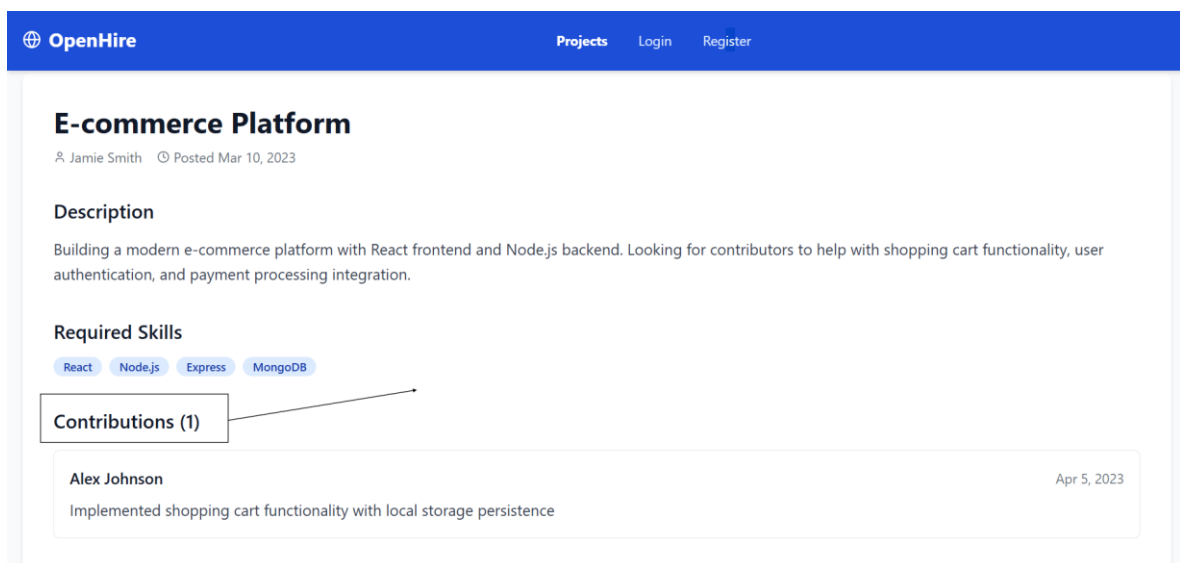
JavaScript, React, Node.js

List technologies and frameworks you're skilled in

Create Account

Already have an account? Login

Allows to choose any user



The project page for 'E-commerce Platform' shows the project title, author (Jamie Smith), and date (Mar 10, 2023). It includes a description, required skills (React, Node.js, Express, MongoDB), and a list of contributions. A callout box highlights the 'Contributions (1)' section.

OpenHire

Projects Login Register

E-commerce Platform

Jamie Smith Posted Mar 10, 2023

Description

Building a modern e-commerce platform with React frontend and Node.js backend. Looking for contributors to help with shopping cart functionality, user authentication, and payment processing integration.

Required Skills

React Node.js Express MongoDB

Contributions (1)

Alex Johnson Apr 5, 2023

Implemented shopping cart functionality with local storage persistence

Project Specific Contributions

CHAPTER 6

CONCLUSIONS AND FUTURE SCOPE

6.1 Conclusions

Open Hire reimagines the open-source contribution journey by transforming it into a structured, rewarding, and career-boosting experience. It not only empowers students and beginners to contribute meaningfully without the overwhelm of traditional platforms, but also ensures their efforts are visible, valued, and impactful. At the same time, it gives maintainers and recruiters a smarter way to discover real talent early—based on verified contributions, not just resumes. By connecting the dots between learning, contributing, and getting hired, Open Hire positions itself as the go-to platform for open-source-driven career growth.

6.1.1 Key Specifications of Open Hire

- **Live Contribution Portfolio:**
The Live Contribution Portfolio on Open Hire captures and showcases each user's real-world coding experience. Every commit, pull request, and resolved issue is automatically tracked and organized by project, complete with details like technologies used, contribution type, and feedback from maintainers. The portfolio serves as a dynamic resume that grows with the user, highlighting measurable impact such as merged PRs and collaboration stats. It's visually designed and shareable, making it easy to present verified contributions to recruiters or peers. This ensures that the user's work doesn't go unnoticed, providing them with continuous visibility and recognition in the open-source and job market.
- **Smart Project Matching**
Open Hire's Smart Project Matching system helps users quickly find relevant open-source projects based on their skills, interests, and experience level. Instead of sifting through thousands of random issues, users are matched with curated, beginner-friendly opportunities. The platform uses AI to analyse tech stacks, issue complexity, documentation quality, and contributor support within projects. It recommends tasks users can confidently handle and grow from. As contributors learn and update their profiles, recommendations evolve to match their development path. This personalized, dynamic approach saves time, reduces confusion, and boosts early engagement by matching users with meaningful projects they're more likely to stick with.
- **Recruiter Dashboard**
The Recruiter Dashboard on Open Hire empowers hiring teams to discover talent based on actual coding contributions instead of traditional resumes. Recruiters can filter candidates by skills, tech stacks, contribution history, and consistency. Each candidate is presented with a detailed, real-time portfolio showing real code, project impact, and collaboration insights. A contribution score helps assess technical and

teamwork abilities at a glance. The dashboard includes communication tools for direct outreach, plus optional filters to promote diversity and inclusion. By offering transparent, skill-based candidate discovery, Open Hire enables recruiters to hire smarter and faster, directly connecting them with proven contributors who are job-ready.

6.1.1 Project Impact

The project has shown positive outcomes in reducing recruitment time and effort while promoting fairness and equal opportunity in hiring. The skill-based matching algorithm improves job fit and minimizes manual shortlisting errors, benefiting both job seekers and employers.

6.1.1.1 Social and Technological Relevance

The system aligns with the growing societal need for inclusive hiring and contributes to the advancement of AI in real-world applications. It also supports the technological shift toward automation in human resource management, making recruitment smarter and more effective.

6.1.1.2 Future Scope

Future developments of *Open Hire* can include expanding the platform to mobile applications, adding features such as real-time video interviews, behavioural analytics, and multilingual support. Integration with third-party job portals, payroll systems, and full HR management tools can turn it into a comprehensive end-to-end hiring ecosystem. Advanced AI capabilities like predictive hiring analytics and cultural fit assessment may also be explored to further enhance decision-making accuracy.

6.2 Vision

OpenHire envisions becoming the global hub for open-source-driven careers, where anyone—regardless of background or location—can build real-world experience, grow their skillset, and get hired based on merit and contribution. By uniting contributors, maintainers, and recruiters in one transparent ecosystem, OpenHire has the power to democratize tech opportunities and redefine how talent is discovered in the 21st century.

APPENDIX

A.2 Questionnaire for users

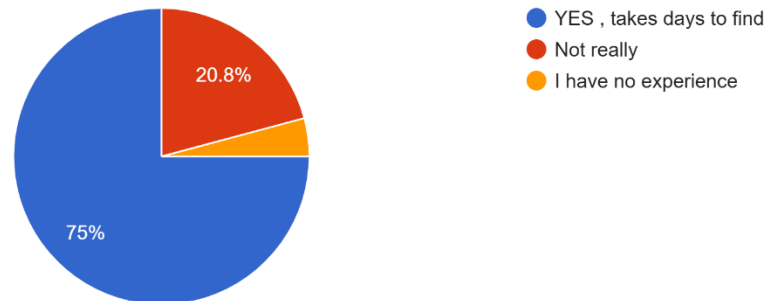
A.1 Questionnaire for User

- 1. Did you ever face problems in finding a project ?**
(Finding projects can be challenging due to lack of guidance; OPENHIRE can streamline access to relevant projects, boosting user engagement and skill-building)
- 2. Did you have experience in contributing to Open - Source Contribution ?**
(Contributing to open-source improves coding skills and collaboration; OPENHIRE can facilitate such contributions, enhancing community growth and user portfolios)
- 3. Do you know about these contributions?**
(Awareness of open-source contributions motivates participation; OPENHIRE can educate users, increasing active involvement and quality contributions)
- 4. Do you feel these freelancing opportunities are required for the present job market?**
(Freelancing offers real-world experience and income; OPENHIRE can connect users to these opportunities, making them more competitive in the job market)
- 5. Do you feel freelancing projects are recognised in this present market of high competitions?**
(Freelancing projects are gaining recognition but still need validation; OPENHIRE can showcase verified projects to improve their credibility and user resumes)
- 6. Would a platform that joins developers and hiring officials would help users and students get hands on-experience and get hired easily?**
(A platform linking developers and employers simplifies hiring and experience-building; OPENHIRE can bridge this gap, accelerating career growth for students and professionals)

A.2 Text Transcripts of User Responses

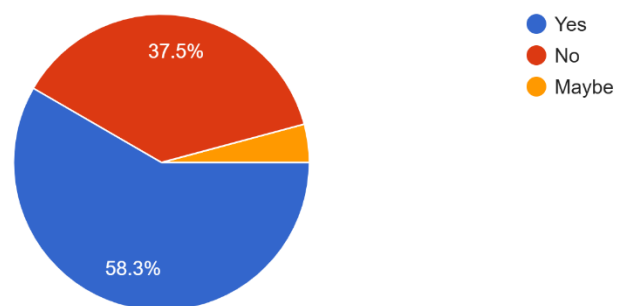
Did you ever face problems in finding a project ?

24 responses



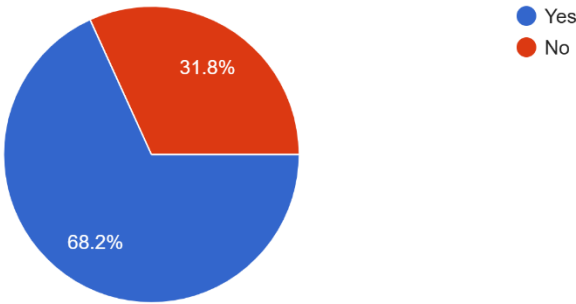
Did you have experience in contributing to Open - Source Contribution ?

24 responses



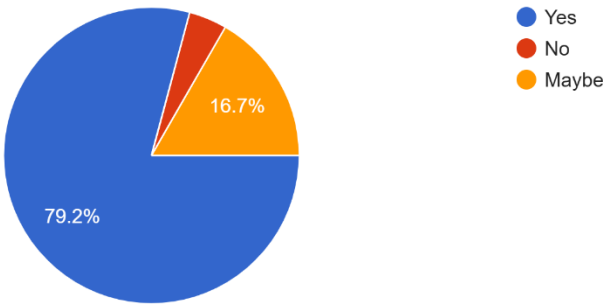
Do you know about these contributions?

22 responses



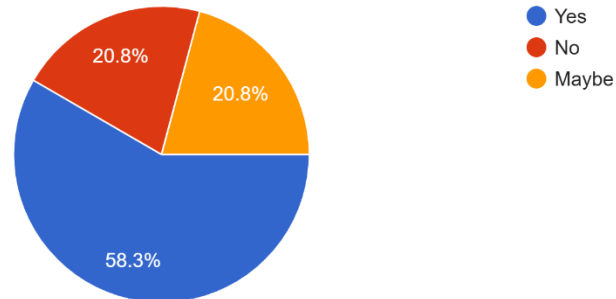
Do you feel these freelancing opportunities are required for the present job market?

24 responses



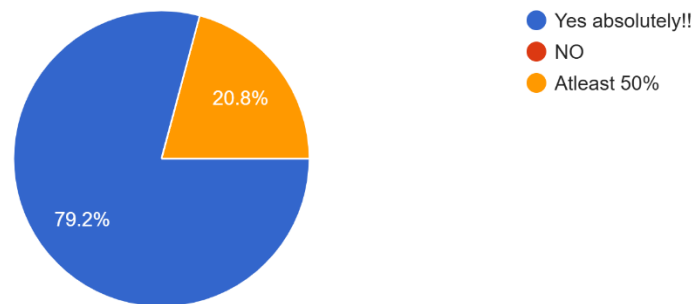
Do you feel freelancing projects are recognised in this present market of high competitions?

24 responses



Would a platform that joins developers and hiring officials would help users and students get hands on-experience and get hired easily?

24 responses



A.3 User Response:

The results suggest that participants generally feel freelancing projects are useful and hold recognition, despite some uncertainty among a minority. This reflects a growing acceptance and acknowledgment of freelancing as a valuable form of work experience in the current market.